ModelIt!

Student Activity Pack

Why Do Magnets Stick?

Exploring Magnetic Forces and Fields

Grade 3 | NGSS 3-PS2-3 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever played with a magnet? What happened when you put it near different things?**

2. **Why do you think magnets stick to the refrigerator but not to a wooden door?**

3. **Draw what you think is happening between a magnet and a paperclip when the magnet pulls it.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Magnet Strength | Magnetic Force | Attraction

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Weak Magnet

Magnet Strength set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Strong Magnet

Magnet Strength set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Medium Magnet

Magnet Strength set to 50%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Distance — How far the magnet is from the metal object — magnetic force gets weaker when the magnet is farther away

Object Material — What the object is made of — only metals like iron, nickel, and cobalt are attracted to magnets

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do magnets stick to some things but not others? Use evidence from your simulation to support your answers.

1 How does magnetic force work even when the magnet is not touching the object?

2 How did your model help you understand why magnets only stick to certain things?

8 STEM Challenge: Design a Magnetic Sorting Machine ModelIt!

YOUR ENGINEERING MISSION

Design a tool that uses magnets to sort metal objects from non-metal objects, like a recycling helper.

THE SCENARIO

A recycling center needs help sorting metal cans from plastic bottles and paper. Your team must design a magnetic sorting tool that can quickly pick up metal items without picking up non-metal items. Use what you learned about magnetic force to make your design!

Engineering Design Thinking

1. WHAT TYPE OF MAGNET WILL YOU USE — A BAR MAGNET, A HORSESHOE MAGNET, OR A DISK MAGNET?

2. HOW WILL YOU ATTACH THE MAGNET TO A HANDLE SO YOU CAN MOVE IT OVER THE PILE?

3. WHAT OBJECTS WILL YOU TEST TO PROVE YOUR SORTER WORKS?

4. HOW WILL YOU MAKE YOUR SORTER WORK FASTER FOR A REALLY BIG PILE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Materials Scientists study what things are made of and why they behave the way they do, including magnetic materials. They earn $60,000-$105,000/year.